

Embedding-DSD

Sentence 1:

the bird flies fast over the hill

Sentence 2:

the car rides fast over the hill

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Sentence 1:

the bird flies fast over the hill

Sentence 2:

the car rides fast over the hill

Base Similarity: 0.64

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Sentence 1:

the **bird** flies fast over the hill

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Sentence 1:

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Base Similarity: 0.64

the car rides fast over the hill

bird car rides fast over the hill

STS: 0.77 | $0.77 - 0.64 \Rightarrow \Delta G = 0.13$

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Sentence 1: the **bird** flies fast over the hill

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Base Similarity: 0.64

the car rides fast over the hill

bird car rides fast over the hill

STS: 0.77 | $0.77 - 0.64 \Rightarrow \Delta G = 0.13$

the **bird** rides fast over the hill

STS: 0.93 | $0.93 - 0.64 \Rightarrow \Delta G = 0.29$

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the car rides fast over the hill

bird car rides fast over the hill

STS: 0.77 | $0.77 - 0.64 \Rightarrow \Delta G = 0.13$

the **bird** rides fast over the hill

STS: 0.93 | $0.93 - 0.64 \Rightarrow \Delta G = 0.29$

the car **bird** fast over the hill

STS: 0.82 | $0.82 - 0.64 \Rightarrow \Delta G = 0.18$

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the **bird** rides fast over the hill

STS: 0.93 | $0.93 - 0.64 \Rightarrow \Delta G = 0.29$

the car **bird** fast over the hill

STS: 0.82 | $0.82 - 0.64 \Rightarrow \Delta G = 0.18$

the car rides **bird** over the hill

STS: 0.74 | $0.74 - 0.64 \Rightarrow \Delta G = 0.10$

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the car **bird** fast over the hill

STS: 0.82 | $0.82 - 0.64 \Rightarrow \Delta G = 0.18$

the car rides **bird** over the hill

STS: 0.74 | $0.74 - 0.64 \Rightarrow \Delta G = 0.10$

the car rides fast **bird** the hill

STS: 0.60 | $0.60 - 0.64 \Rightarrow \Delta G = -0.04$

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the car **bird** fast over the hill

STS: 0.82 | $0.82 - 0.64 \Rightarrow \Delta G = 0.18$

the car rides **bird** over the hill

STS: 0.74 | $0.74 - 0.64 \Rightarrow \Delta G = 0.10$

the car rides fast **bird** the hill

STS: 0.60 | $0.60 - 0.64 \Rightarrow \Delta G = -0.04$

the car rides fast over **bird** hill

STS: 0.73 | $0.73 - 0.64 \Rightarrow \Delta G = 0.09$

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the car rides **bird** over the hill

STS: 0.74 | $0.74 - 0.64 \Rightarrow \Delta G = 0.10$

the car rides fast **bird** the hill

STS: 0.60 | $0.60 - 0.64 \Rightarrow \Delta G = -0.04$

the car rides fast over **bird** hill

STS: 0.73 | $0.73 - 0.64 \Rightarrow \Delta G = 0.09$

the car rides fast over the **bird**

STS: 0.73 | $0.64 - 0.64 \Rightarrow \Delta G = 0.64$

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STS: 0.82 | $0.82 - 0.64 \Rightarrow \Delta G = 0.18$

the car rides bird over the hill

STS: 0.74 | $0.74 - 0.64 \Rightarrow \Delta G = 0.10$

the car rides fast bird the hill

STS: 0.60 | $0.60 - 0.64 \Rightarrow \Delta G = -0.04$

the car rides fast over bird hill

STS: 0.73 | $0.73 - 0.64 \Rightarrow \Delta G = 0.09$

the car rides fast over the bird

STS: 0.73 | $0.64 - 0.64 \Rightarrow \Delta G = 0.64$

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...

Embedding-DSD

`the` → [0.13, 0.29, 0.09, 0.10, ..., 0.02]

Embedding-DSD

`the` → [0.13, 0.29, 0.09, 0.10, ..., 0.02]

`car` → [0.29, 0.29, 0.18, 0.15, ..., 0.09]

Embedding-DSD

the → [0.13, 0.29, 0.09, 0.10, ..., 0.02]

car → [0.29, 0.29, 0.18, 0.15, ..., 0.09]

rides → [0.18, 0.24, 0.17, 0.12, ..., 0.08]

Embedding-DSD

the → [0.13, 0.29, 0.09, 0.10, ..., 0.02]

car → [0.29, 0.29, 0.18, 0.15, ..., 0.09]

rides → [0.18, 0.24, 0.17, 0.12, ..., 0.08]

...

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$$\text{AggrGain}_{\text{unigram}} = \frac{1}{n} \cdot \sum_{i=1}^n \frac{\text{gains}_i}{i}$$